

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

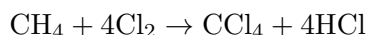
Paper No. CHB-02A: Ancillary Chemistry-I

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Indicate the reducing and oxidizing agents in the following reaction:



- (b) Write the composition of stainless steel.
- (c) What is slag? Write its chemical formula.
- (d) Write the names and formulae of two important ores of iron.
- (e) Define the term 'isoelectric point' of an amino acid.

Question 2. Differentiate between the following pairs: [7 × 2 = 14]

- (a) Distillation and Fractional distillation
- (b) TLC (Thin Layer Chromatography) and Paper Chromatography
- (c) Primary structure of proteins and secondary structure
- (d) Melting point as a purity criterion of a crystalline compound

Question 3. Comment upon the role of metals in our life. How is iron recovered from its ores? Write the chemical reactions involved in the blast furnace. Explain the terms: pig iron, cast iron, and wrought iron. [14]

Question 4. Explain the following giving chemical reactions involved: ... [4 + 4 + 4 + 2 = 14]

- (a) Preparation and analytical applications of Fehling's solution
- (b) Preparation and analytical applications of Benedict's reagent
- (c) Determination of glucose in urine using Tollens' reagent
- (d) What is the role of Cobalt in the human body?

Question 5. Why are enzymes also called biocatalysts? Describe the different types of enzymes. What is a cofactor? Describe the denaturation of enzymes in very brief. [14]

Question 6.

(a) Write short notes on: [4 × 2 = 8]

(i) Mutarotation

(ii) Zwitterion

(b) Discuss the structure of Nylon-6,6 and Buna-S rubber. Give their monomer units. [6]